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Online Learning with Relative Lipschitz Losses

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January 7, 2021

Outline

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2. Classic Results for Lipschitz Losses
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Online Convex Optimization

Online Convex Optimization: Introduction

Algorithm: Online Convex Optimization

Input: A convex set S.

for $t = 1, \ldots, T$ **do**

  Predicts a vector $\mathbf{w}_t \in S$.

  Recieve a convex loss function $l_t : S \to \mathbb{R}$.

  Suffer loss $l_t(\mathbf{w}_t)$.

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$$\text{Regret}_T(\mathbf{u}) = \sum_{t=1}^{T} f_t(\mathbf{w}_t) - \sum_{t=1}^{T} f_t(\mathbf{u}).$$

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- Our goal is to minimize:

$$\max_{\mathbf{u} \in S} \Big(\text{Regret}_T(\mathbf{u}) \Big)$$

Online Convex Optimization: Applications

- **Convex Optimization:** Adversary plays a fixed f.

$$f\left(\frac{1}{T}\sum_{t=1}^{T}\mathbf{w}_t\right) - f(\mathbf{w}*) \leq \frac{1}{T}\sum_{t=1}^{T} f(\mathbf{w}_t) - \frac{1}{T}\sum_{t=1}^{T} f(\mathbf{w}^*) \leq \frac{\mathrm{Regret}_T(\mathbf{w}^*)}{T}$$

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- **Online Linear Regression**:
  - The Learner Receives $\mathbf{x}_t$.
  - The Learner Decides $\mathbf{w}_t$.
  - The Adversary Reveals y_t.
  - The loss $l_t = (\langle \mathbf{w}_t, \mathbf{x}_t \rangle - y_t)^2$ is suffered.

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- **Other Applications**: Online Classification, Online Pricing, Online Ranking, Expert Advice, ...

Classic Results for Lipschitz Losses

Achievability: Online Mirror Descent

Algorithm: Online Mirror Descent

for $t = 1, \ldots, T$ **do**

Choose $\mathbf{w}_t$.

Recieve $l_t : \mathbb{R}^d \to (-\infty, \infty]$ and pay $l_t(\mathbf{w}_t)$.

Set $\mathbf{g}_t \in \partial l_t(\mathbf{w}_t)$

Set $\mathbf{w}_{t+1} = \arg\min_{\mathbf{w} \in S} \langle \mathbf{g}_t, \mathbf{w} \rangle + \frac{1}{\eta_t} B_\psi(\mathbf{w} || \mathbf{w}_t)$

end

Regret Bound [Abernethy et al., COLT 2014]

Assume that ψ is $\lambda-$ strongly convex with respect to $||.||$, S is a closed convex set, and $||\mathbf{g}||_\star^2 \leq L$. OMD with $\eta_t = \frac{C\sqrt{\lambda}}{\sqrt{\sum_{i=1}^t ||\mathbf{g}||_\star^2}}$ achieves:

$$\text{Regret}(\mathbf{u}) \leq \frac{2CL}{\sqrt{\lambda}} \sqrt{T}.$$

Converse Bounds

Lower Bounds Classic Results: e.g. [Orabona, 2021]

Let S be any bounded convex set, $D = \text{diam}(S)$, and $\mathcal{A}$ be any algorithm for Online Linear Optimization on S. There exists a sequence $\mathbf{g}_1, \ldots, \mathbf{g}_T$ with $||\mathbf{g}_t||_2 \leq L$ such that

$$\exists \mathbf{u} \in S: \quad \text{Regret}_{\text{T}}(\mathbf{u}) = \sum_{t=1}^{T} \langle \mathbf{w}_t - \mathbf{u}, \mathbf{g}_t \rangle \geq \frac{\sqrt{2} L D \sqrt{T}}{4}.$$

Relative Lipschitz Functions

Relative Lipschitz Functions

Definitions

A function $f : S \to \mathbb{R}$ is said to be **Lipschitz relative to function** R if

$$\forall \mathbf{x}, \mathbf{y} \in S, \forall \mathbf{g} \in \partial f(\mathbf{x}) : ||\mathbf{g}||_*^2 \leq L^2 \frac{B_R(\mathbf{y}||\mathbf{x})}{\frac{1}{2}||\mathbf{y} - \mathbf{x}||_2^2}$$

- For example, the function $f(x) = x^2$ is not Lipschitz on $\mathbb{R}$. However, it is f is $\sqrt{2}$-Lipschitz relative to $R(x) = 2x^4$.

Relative Lipschitz Functions in Online Learning

Relative Lipschitzness in Online Learning

Relative Lipschitz Property was introduced in [Lu et al., SIAM Opt., 2018] for Convex Optimization.

Relative Lipshitzness FTRL

If the functions f_t is L-Lipschitz relative to R.

- [Zhou et al., NeurIPS, 2020]
  The FTRL algorithm, $\mathbf{w}_{t+1} = \arg\min_{\mathbf{w}\in S} \left(\sum_{i=1}^{t} f_t(\mathbf{w}) + \frac{1}{\eta_t} R(\mathbf{w}) \right)$, achieves
  $$\text{Regret}_T(\mathbf{u}) \leq \frac{K}{\eta_T} + \sum_{t=1}^{T} \frac{L^2 \eta_{t-1}}{2}.$$
- [B. M., 2020]
  The FTRL algorithm, $\mathbf{w}_{t+1} = \arg\min_{\mathbf{w}\in S} \left(\sum_{i=1}^{t} f_t(\mathbf{w}) + \frac{1}{\eta_t} \psi(\mathbf{w}) \right)$, achieves
  $$\text{Regret}_T(\mathbf{u}) \leq \frac{K}{\eta_T} + \sum_{t=1}^{T} \frac{L^2 \eta_{t-1}}{2} + \sum_{t=1}^{T} \frac{B_\epsilon(\mathbf{x}_{t+1} || \mathbf{x}_t)}{\eta_{t-1}}.$$

  where $\epsilon(.) = R(.) - \psi(.)$.

Relative Lipschitzness in Online Learning

Relative Lipshitzness OMD [B. M., 2020]

If the functions f_t is L-Lipschitz relative to R, and we run Online Mirror Descent with η_t and link function ψ, we have

$$\text{Regret}_T(\mathbf{u}) = \frac{K}{\eta_T} + \sum_{t=1}^{T} \frac{\eta_t L^2}{2} + \sum_{t=1}^{T} \frac{B_\epsilon(\mathbf{x}_{t+1}||\mathbf{x}_t)}{\eta_t},$$

where $\epsilon(.) = R(.) - \psi(.)$.

Ongoing Work

- Online Convex Optimization with Sub-Gaussian Noise:
  - The learner has access to a noisy version of the loss function or sub-gradient.
  - How does the tail bound affect the Expected Regret?
  - A similar problem was studied in [Jun et al., COLT, 2019] for Parameter-Free Online Learning.

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- Parameter-Free Online Learning with Relative Lipschitz Losses:
  - Setting η_t requires the knowledge of L and R.
  - Rich literature on parameter-free online learning: [Cutkosky, COLT 2020], [Orabona, NeurIPS, 2016].

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- Lower Bounds: Classic techniques cannot be easily employed.

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